THE UK UNIVERSITY INTEGRATION BEE 2025/2026

$$\int$$

ROUND TWO

GROUP ROUND

Sponsored by



Jane Street

Problems

1. $\displaystyle\int_0^{\pi} \ln\left(\cos(x) + \sqrt{1+\cos^2(x)}\right) \mathrm{d}x$
2. $\displaystyle\int_0^{1} \frac{1}{\sqrt[3]{\sqrt[3]{x} - x}} \mathrm{d}x$
3. $\displaystyle\int_0^{1} \frac{2x-1}{\sqrt{2x-x^2} - \sqrt{1-x^2}} \mathrm{d}x$
4. $\displaystyle\int_1^{\infty} \{x\} \left\lfloor \frac{2}{\{x\}} \right\rfloor^{-\lfloor x \rfloor} \mathrm{d}x$
5. $\displaystyle\int_0^{\pi/2} (\sin(x) + \cos(x)) \tan^{-1}(\sin(x)) \, \mathrm{d}x$
6. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos^2(x)}{\cos^2(x) + (x\cos(x) + \sin(x) + 1)^2} \mathrm{d}x$
7. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(e+e^x)(x^4 - 4x^3 + 6x^2 - 4x + 2)} \mathrm{d}x$
8. $\displaystyle\int_0^{\infty} \frac{\sin^6(x)}{x^2} + \frac{\sin^5(x)}{x} + \frac{\sin^4(x)}{x^2} + \frac{\sin^3(x)}{x^3} \mathrm{d}x$
9. $\displaystyle\sum_{n=1}^{\infty} \int_0^1 (-1)^n \ln\left(\frac{1 - x^{\frac{1}{n}}}{1 - x^{\frac{1}{n+1}}}\right) \mathrm{d}x$
10. $\displaystyle\left\lceil \int_1^{\infty} \frac{1}{\tan^{-1}(x)} - \frac{1+x^2}{1 + \tan^{-1}(x) + x^2\tan^{-1}(x)} \mathrm{d}x \right\rceil$